Name:- Muhammad Obaid
Roll.No:- CT-23025
Class:- CSIT-A

# Probability & Statistics (MT-331) Assignment

## Exercise 5

## Q 5.7

Given:-

$p = 70\% = 0.7$ (chain-smokers)

$q = 0.3$ ; $n = 10$ patient

Find:-

a) $p(x < 5) = ?$

b) $n = 20$ ; $p(x < 10) = ?$

Sol:-

a) $p(x < 5) = p(x \leq 4)$

Then a/c to binomial distribution:-

$$p(x \leq 4) = \sum_{x=0}^{4} b(x; 10, 0.7)$$

$$= \sum_{x=0}^{4} \left[ {}^{10}C_x (0.7)^x (0.3)^{10-x} \right]$$

$$p(x \leq 4) = 0.0473$$

b) $p(x < 10) = p(x \leq 9)$

So by binomial distribution:-

$$p(x \leq 9) = \sum_{x=0}^{9} b(x; 20, 0.7)$$

$$= \sum_{x=0}^{9} \left[ {}^{20}C_x (0.7)^x (0.3)^{20-x} \right]$$

$$p(x \leq 9) = 0.0171$$

## Q 5.8

Given:-

$p = 0.6$ (Took valium), $q = 0.4$

$n = 8$ users

Find:-

a) $p(x=3) = ?$

b) Atleast 5-users taking Valium were not psychological = ?

$\{p(x \leq 3)\}$ or $\therefore p(x \geq 5)$ if $p = 0.4$ (Taken)

Sol:-

a) $p(x=3) = b(3; 8, 0.6) = {}^{8}C_3 (0.6)^3 (0.4)^5$

$p(x=3) = 0.124$

b) $p$ (Atleast 5-users taking Valium were not psychological)

$= p$ (Atmost 3-users taking valium that were psychological) $\Rightarrow p(x \leq 3)$

$p(x \leq 3) = b(x; 3, 0.6) = \sum_{x=0}^{3} \left( {}^{8}C_x (0.6)^x (0.4)^{8-x} \right)$

$p(x \leq 3) = 0.174$

## Q 5.9

Given:-

$p = 25\% = 0.25$ (have blowouts)

$q = 0.75$

$n = 15$

Find:-

a) $p(3 \leq x \leq 6) = ?$

b) $p(x < 4) = ?$

c) $p(x > 5) = ?$

Sol:-

a) $p(3 \leq x \leq 6) = b(x; 15, 0.25) = \sum_{x=3}^{6} \left( {}^{15}C_x (0.25)^x (0.75)^{15-x} \right)$

$p(3 \leq x \leq 6) = 0.707$

b) $p(x < 4) = \sum_{x=0}^{3} \left( {}^{15}C_x (0.25)^x (0.75)^{15-x} \right)$

$p(x < 4) = 0.461$

c) $p(x>5) = 1 - p(x \leq 5)$

$= 1 - \sum_{x=0}^{5} b(x; 15, 0.25)$

$p(x>5) = 0.148$

## Q 5.10

Given:-

$p = 70\% = 0.7$ (disapprove pot-smoking)

$q = 0.3$

$n = 12$

Find:-

a) $p(7 \leq x \leq 9) = ?$

b) $p(x \leq 5) = ?$

c) $p(x \geq 8) = ?$

OR can do like $p(x \leq 9) - p(x \leq 6)$

$\sum_{x=0}^{9} - \sum_{x=0}^{6}$

Sol:-

a) $p(7 \leq x \leq 9) = \sum_{x=7}^{9} b(x; 12, 0.7) = \sum_{x=7}^{9} \left[ {}^{12}C_x (0.7)^x (0.3)^{12-x} \right]$

$p(7 \leq x \leq 9) = 0.629$

b) $p(x \leq 5) = \sum_{x=0}^{5} b(x; 12, 0.7)$

$p(x \leq 5) = 0.0386$

c) $p(x \geq 8) = \sum_{x=8}^{12} b(x; 12, 0.7)$

$p(x \geq 8) = 0.724$

## Q 5.11

Given:-

$p = 0.9$ (recovers), $q = 0.1$

$n = 7$

$p(x=5) = ?$

Sol:-

$p(x=5) = b(5; 7, 0.9)$

$p(x=5) = {}^{7}C_5 (0.9)^5 (0.1)^2$

$p(x=5) = 0.124$

## Q 5.12

Given:-

$q = 75\% = 0.75$ (within-state)

$p = 0.25$

$n = 9$

$p(x<4) = ?$

Sol:-

$$p(x<4) = \sum_{x=0}^{3} b(x; 9, 0.25) = \sum_{x=0}^{3} \left[ {}^{9}C_{x} (0.25)^{x} (0.75)^{9-x} \right]$$

$p(x<4) = 0.834$

## Q 5.15

Given:-

$q = 0.6$ (protected i.e. non-contracted from disease)

$p = 0.4$

$n = 5$

Find:-

a) $p(x=0) = ?$ b) $p(x<2) = ?$ c) $p(x>3) = ?$

Sol:-

a) $p(x=0) = b(0; 5, 0.4) = {}^{5}C_{0} (0.4)^{0} (0.6)^{5}$

$p(x=0) = 0.0778$

b) $$p(x<2) = \sum_{x=0}^{1} b(x; 5, 0.4) = \sum_{x=0}^{1} \left[ {}^{5}C_{x} (0.4)^{x} (0.6)^{5-x} \right]$$

$p(x<2) = 0.337$

c) $$p(x>3) = \sum_{x=4}^{5} b(x; 5, 0.4)$$

$p(x>3) = 0.08704$

## Q 5.16

Given:- independently → with replacement

$p = 0.6$ (engine not fail)

$q = 0.4$

⇒ Plane can make a safe flight if atleast one-half of its engine run = ?

Find:-

Which engine has higher probability for successful flight = ?

(4-engine plane) OR (2-engine plane)

Sol:-

For safe flight of 4-engine plane:- $n=4$

$\Rightarrow p(x \geq 2) = \sum_{x=2}^{4} b(x;4,0.6) = \sum_{x=2}^{4} \left[ {}^4C_x (0.6)^x (0.4)^{4-x} \right]$

$p(x \geq 2) = 0.8208$

For safe flight of 2-engine plane:- $n=2$

$\Rightarrow p(x \geq 1) = 1 - p(x=0)$

$p(x \geq 1) = 1 - \left[ {}^2C_0 (0.6)^0 (0.4)^2 \right] = 0.84$

Since $0.84 > 0.8208$

so, 2-engine plane has higher successful flight probability than 4-engine plane.

## Q 5.29

Given:- By default → (without replacement)

$N = 5$ (tulips) $+ 4$ (daffodils) $= 9$

$n = 6$

$k = 5$, $N-k = 4$

Plant 2 daffodils and 4 tulips i.e. $p(x=4) = ?$ → as k taken was for tulip

Sol:- By hypergeometric distribution:- $h(x;N,n,k) = \dfrac{{}^kC_x \; {}^{N-k}C_{n-x}}{{}^NC_n}$

$p(x=4) = h(4;9,6,5) = \dfrac{{}^5C_4 \; {}^4C_2}{{}^9C_6}$

$p(x=4) = \frac{5}{14} = 0.357$

## Q 5.30

Given:-

$N = 9$ (vitamins) $+ 6$ (narcotic tablets) $= 15$

$n = 3$

Probability that traveller will be arrested: $p(x \geq 1) = ?$

$k = 6$, $N-k = 9$

Sol:-

$p(x \geq 1) = 1 - p(x=0) = 1 - h(0;15,3,6)$

$p(x \geq 1) = 1 - \dfrac{{}^6C_0 \; {}^9C_3}{{}^{15}C_3} = 0.815$

# Q 5.31

Given:-

$N = 4(\text{doctors}) + 2(\text{nurses}) = 6$

$n = 3$

Formula for distributing representing doctors = ?

$k = 4,\ N-k = 2$

$p(2 \leq x \leq 3) = ?$

Sol:-

$$h(x; 6, 3, 4) = \frac{{}^4C_x \ {}^2C_{3-x}}{{}^6C_3}$$

For $p(2 \leq x \leq 3)$:-

$p(2 \leq x \leq 3) = 1 - p(x=1)$ → As committ can be atmax of 3.

$$= 1 - \frac{{}^4C_1 \ {}^2C_2}{{}^6C_3}$$

$p(2 \leq x \leq 3) = 0.8$

# Q 5.32

Given:-

$N = 10$ missiles

$n = 4$

$k = 3$ (defective)

Find:-

OR can take $k=3$; $x=0$ → As no defective will fire.

a) All missiles fire so $k=7$; $p(x=4) = ?$

b) Atmost 2 not fire $p(x \leq 2) = ?$

Sol:-

a) $$^{k=7}p(x=4) = h(4; 10, 4, 7) = \frac{{}^7C_4 \ {}^3C_0}{{}^{10}C_4}$$

$p(x=4) = 0.167$

b) $$^{k=3}p(x \leq 2) = \sum_{x=0}^{2} h(x; 10, 4, 3) = \frac{{}^3C_x \ {}^7C_{4-x}}{{}^{10}C_4}$$

$p(x \leq 2) = 0.967$

## Q 5.33

Given:-

$N = 52$ cards

$n = 7$

Find:-

a) Exactly 2 are face-cards; $k = 12$ $p(x=2) = ?$

b) Atleast 1 is queen ; $k = 4$ $p(x \geq 1) = ?$

Sol:-

a) $p(x=2) = h(2; 52, 7, 12) = \dfrac{{}^{12}C_2 \; {}^{40}C_5}{{}^{52}C_7}$

$p(x=2) = 0.325$

b) $p(x \geq 1) = 1 - p(x=0)$ → OR can do $p(1 \leq x \leq 4)$ as at max queens can be 4.

$= 1 - h(0; 52, 7, 4) = 1 - \dfrac{{}^{4}C_0 \; {}^{48}C_7}{{}^{52}C_7}$

$p(x \geq 1) = 0.4497$

## Q 5.36

Given:- Sent-back → without replacement for screening

$N = 25$ boxes

$n = 3$

Find:-

a) Box containing '3' defectives is shipped $k = 3$ ; $p(x=0) = ?$

b) Box containing only 1 defective is shipped ; $k = 1$ $p(x=1) = ?$

Sol:-

a) $p(x=0) = h(0; 25, 3, 3) = \dfrac{{}^{3}C_0 \; {}^{22}C_3}{{}^{25}C_3}$

$p(x=0) = 0.6696$

b) $p(x=1) = h(1; 25, 3, 1) = \dfrac{{}^{1}C_1 \; {}^{24}C_2}{{}^{25}C_3}$

$p(x=1) = 0.12$

## Q 5.37

Given:- Inspector "replaces" defective item.

$N = 25$ boxes

$n = 3$

Find:-

a) $p(x=0) = ?$ $p = \frac{3}{25}$, $q = \frac{22}{25}$

b) ~~$p(x=1)$~~ $p(1 \leq x \leq 3) = ?$ Defective can be in any box as replacement is allowed

$p = \frac{1}{25}$, $q = 24/25$

Sol:-

a) A/c to Binomial distribution:-

$$b(x; n, p) = {}^{n}C_{x}\, p^{x}\, q^{n-x}$$

$$p(x=0) = b\left(0; 3, \frac{3}{25}\right) = {}^{3}C_{0}\left(\frac{3}{25}\right)^{0}\left(\frac{22}{25}\right)^{3}$$

$$p(x=0) = 0.681$$

b) $p(x \geq 1) = 1 - p(x=0) = 1 - b\left(0; 3, \frac{1}{25}\right)$

$$p(x \geq 1) = 1 - \left[{}^{3}C_{0}\left(\frac{1}{25}\right)^{0}\left(\frac{24}{25}\right)^{3}\right]$$

$$p(x \geq 1) = 0.115$$

## Q 5.40

Given:- As 1 person can vote 1 time only → No replacement

$N = 10{,}000$

$n = 15$

Atmost 7 person favour tax; $k = 6000$ $p(x \leq 7) = ?$

Sol:-

$$p(x \leq 7) = \sum_{x=0}^{7} h(x; 10000, 15, 6000)$$

$$p(x \leq 7) = \frac{{}^{6000}C_{x}\ {}^{4000}C_{15-x}}{{}^{10000}C_{15}}$$

$$p(x \leq 7) = 0.213$$

# Q 5.42

Given:-

$N = 52$

$n = 13$ (bridge hand distribution)
multiple orders

$a_1 = a_2 = a_3 = a_4 = 13$

$x_1 = 5$ spades, $x_2 = 2$ hearts, $x_3 = 3$ diamonds, $x_4 = 3$ clubs

Probability of card distribution = ?

Sol:-

A/c to multivariate Hypergeometric distribution:-

$$f(x_1, x_2, \ldots x_n; a_1, a_2, \ldots a_n; N, n) = \frac{{}^{a_1}C_{x_1}\ {}^{a_2}C_{x_2} \cdots {}^{a_k}C_{x_k}}{{}^{N}C_n}$$

$$f(5,2,3,3; 13,13,13,13; 52,13) = \frac{{}^{13}C_5\ {}^{13}C_2\ {}^{13}C_3\ {}^{13}C_3}{{}^{52}C_{13}}$$

Probability $= 0.0129$

# Q 5.43

Given:-

$N = 2$ (canadians) $+ 3$ (Japanese) $+ 5$ (Italians) $+ 2$ Germans $= 12$

$n = 4$

$a_1 = 2, a_2 = 3, a_3 = 5, a_4 = 2$

Find:-

a) All nationalities are represented $(x_1 = x_2 = x_3 = x_4 = 1) = ?$

b) All are present except Italian = ?

Sol:-

a) $$p(A) = \frac{{}^{2}C_1\ {}^{3}C_1\ {}^{5}C_1\ {}^{2}C_1}{{}^{12}C_4} = 0.121$$

b) $p(B) = p(x_1 = 2, x_2 = 1, x_3 = 0, x_4 = 1) + p(1,2,0,1) + p(1,1,0,2)$

$$p(B) = \frac{{}^{2}C_2\ {}^{3}C_1\ {}^{5}C_0\ {}^{2}C_1}{{}^{12}C_4} + \frac{{}^{2}C_1\ {}^{3}C_2\ {}^{5}C_0\ {}^{2}C_1}{{}^{12}C_4} + \frac{{}^{2}C_1\ {}^{3}C_1\ {}^{5}C_0\ {}^{2}C_2}{{}^{12}C_4}$$

$p(B) = 0.0485$

## Q 5.44

Given:-

$N = 9$

$n = 5$

$a_1 = 3$ Green, $a_2 = 2$ blue, $a_3 = 4$ Red

Probability that both blue balls and atleast 1 Red is selected = ?

Sol:-

$$p(A) = P(x_1 = 2, x_2 = 2, x_3 = 1) + P(1,2,2) + P(0,2,3)$$

$$p(A) = \frac{{}^3C_2\ {}^2C_2\ {}^4C_1}{{}^9C_5} + \frac{{}^3C_1\ {}^2C_2\ {}^4C_2}{{}^9C_5} + \frac{{}^3C_0\ {}^2C_2\ {}^4C_3}{{}^9C_5}$$

$$p(A) = \frac{34}{126} = 0.2698$$

OR

$$p(A) = \frac{{}^2C_2}{{}^9C_5} \sum_{x=1}^{3} \left({}^3C_{3-x}\ {}^4C_x\right) = 0.2698$$

## Q 5.50

Given:-

$p = \frac{1}{2}$, $q = \frac{1}{2}$

Find:-

a) Probability of $3^{rd}$ Head on $7^{th}$ flip = $p(x=7)$ = ?

b) $1^{st}$ Head on $4^{th}$ flip : $p(x=4)$ = ?

Sol:-

a) A/c to Negative Binomial distribution:-

$$b^*(x; k, p) = {}^{x-1}C_{x-1}\ p^k q^{x-k}$$

$$p(x=7) = b^*\left(7; 3, \frac{1}{2}\right) = {}^6C_2 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^4$$

$$p(x=7) = 0.117$$

b) For $1^{st}$ success after '$i$' tries; using Geometric distribution:-

$$p(x=4) = g\left(4; \frac{1}{2}\right) = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)^3$$

$$p(x=4) = 0.0625$$

## Q 5.51

Given:-

$n = 3$, $p = \frac{6}{8} = \frac{3}{4}$ (For odd one), $q = \frac{2}{8} = \frac{1}{4}$ ({T,T,T}, {H,H,H})

Find:-

$P(1 \leq x \leq 3) = ?$

Sol:-

Since only single odd coin is required so using Geometric distribution.

$$P(x \leq 3) = \sum_{x=1}^{3} pq^{x-1} = \frac{3}{4} \sum_{x=1}^{3} \left(\frac{1}{4}\right)^{x-1}$$

$$P(x \leq 3) = 0.984$$

## Q 5.53

Given:-

$\mu = 5$ times /day

Find:-

a) $P(x > 5) = ?$

b) $P(x = 0) = ?$

Sol:-

a) For average interval, using Poisson distribution:-

$$P(x > 5) = 1 - P(x \leq 5) \cancel{= 1 - P(x; 5)}$$

$$= 1 - \sum_{x=0}^{5} P(x; 5)$$

$$P(x > 5) = 1 - \sum_{x=0}^{5} \frac{e^{-5} 5^{x}}{x!} = 1 - \left(e^{-5} \sum_{x=0}^{5} \frac{5^{x}}{x!}\right)$$

$$P(x > 5) = 0.384$$

b) $P(x = 0) = P(0; 5)$

$$P(x = 0) = \frac{e^{-5} 5^{0}}{0!}$$

$$P(x = 0) = 0.00674$$

## Q 5.55

Given:-

$p = 0.7$

$q = 0.3$

Find:-

a) Student will pass test on $3^{rd}$ try : $p(x=3) = ?$

b) Before $4^{th}$ try : $p(x \leq 3) = ?$

Sol:-

a) A/c to Geometric Distribution:-

$$p(x=3) = (0.7)(0.3)^{3-1} = 0.063$$

b) $$p(x \leq 3) = \sum_{x=1}^{3} bg(x; 0.7) = p \sum_{x=1}^{3} q^{x-1}$$

$$p(x \leq 3) = (0.7) \sum_{x=1}^{3} (0.3)^{x-1}$$

$$p(x \leq 3) = 0.973$$

## Q5.56

Given:-

$\mu = 3$ accidents / month

Find:?

a) $p(x=5) = ?$

b) $p(x<3) = ?$

c) $p(x \geq 2) = ?$

Sol:-

a) By Poisson Distribution:-

$$p(x=5) = \frac{e^{-3}(3)^5}{5!} = 0.1008$$

b) $$p(x \leq 2) = \sum_{x=0}^{2} p(x;3) = e^{-3} \sum_{x=0}^{2} \frac{3^x}{x!}$$

$$p(x \leq 2) = 0.423$$

c) $$p(x \geq 2) = 1 - p(x<2)$$

$$= 1 - \left(e^{-3} \sum_{x=0}^{1} \frac{3^x}{x!}\right)$$

$$p(x \geq 2) = 0.8009$$

## Q 5.57

Given:-

$\mu = 2$ errors / page

Find:-

a) $p(x \geq 4) = ?$ b) $p(x=0) = ?$

Sol:-

a) By Poisson distribution:-

$$p(x \geq 4) = 1 - p(x \leq 3)$$

$$= 1 - \left(e^{-2} \sum_{x=0}^{3} \frac{2^x}{x!}\right)$$

$$p(x \geq 4) = 0.143$$

b) $p(x=0) = p(0;2) = \frac{e^{-2}(2)^0}{0!}$

$$p(x=0) = 0.135$$

## Q 5.58

Given:-

$\mu = 6$ hurricanes / year

Find:-

a) $p(x < 4) = ?$ b) $p(6 \leq x \leq 8) = ?$

Sol:-

a) By Poisson distribution:-

$$p(x \leq 3) = e^{-6} \sum_{x=0}^{3} \frac{6^x}{x!}$$

$$p(x \leq 3) = 0.151$$

b) $p(6 \leq x \leq 8) = p(x=6) + p(x=7) + p(x=8)$

$$= e^{-6}\left[\frac{6^6}{6!} + \frac{6^7}{7!} + \frac{6^8}{8!}\right]$$

$$p(6 \leq x \leq 8) = 0.402$$

## Q 5.59

Given:-

$p = 0.8$ (believe tale)

$q = 0.2$

Find:-

a) $6^{th}$ person to hear tale is $4^{th}$ one to believe; $p(x=6) = ?$

b) $3^{rd}$ person is $1^{st}$ one to believe; $p(x=3) = ?$

Sol:-

a) By Negative Binomial distribution:-

$p(x=6) = b^*(6, 4, 0.8) = {}^5C_3 (0.8)^4 (0.2)^2$

$p(x=6) = 0.164$

b) By Geometric distribution:-

$p(x=3) = g(3; 0.8) = (0.8)(0.2)^{3-1}$

$p(x=3) = 0.032$

Q 5.60

Given:-

$\mu = 12$ mice / acre

Find:-

a) $p(x<7) = ?$

b) Fewer than 7 mice found on 2 of next 3 acres ; $p(x=2) = ?$

Sol:-

a) By Poisson Distribution:-

$p(x \leq 6) = e^{-12} \sum_{x=0}^{6} \frac{12^x}{x!}$

$p(x \leq 6) = 0.0458$

b) Now apply Binomial distribution of $p = 0.04582$, $n = 3$, $x = 2$

$p(x=2) = b(2; 3, 0.04582) = {}^3C_2 (0.04582)^2 (1-0.04582)^{3-2}$

$p(x=2) = 0.00601.$

Q 5.61

Given:-

$p = \frac{1}{1000}$ error (directly on average probability is given)

$n = 10,000$

$p(6 \leq x \leq 8) = ?$ → ORed $p(x=6) + p(x=7) + p(x=8)$

Sol:-

Using binomial distribution:-

$p(6 \leq x \leq 8) = \sum_{x=6}^{8} {}^{10000}C_x (0.001)^x (1-0.001)^{10,000-x}$

$p(6 \leq x \leq 8) = 0.266$

OR since p(average) was given so $\mu = \frac{1}{1000} \times 10,000 = 10$ error/person

$p(6 \leq x \leq 8) = e^{-10} \left[ \frac{10^6}{6!} + \frac{10^7}{7!} + \frac{10^8}{8!} \right] = 0.266$